CMSI 3802: Languages & Automata II

Homework #1 due January 31, 2025

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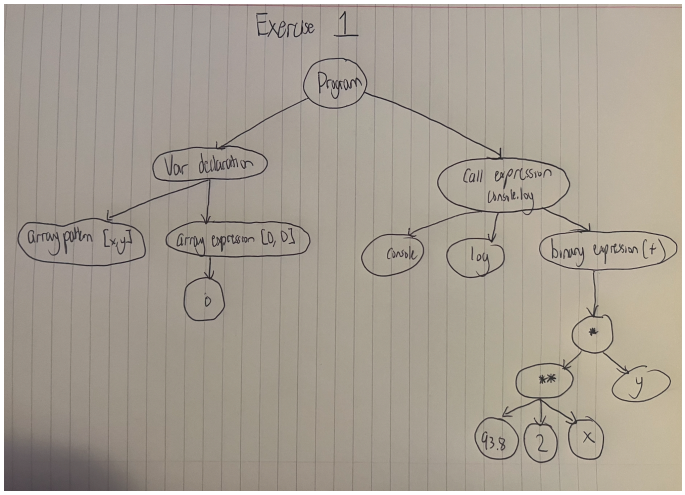
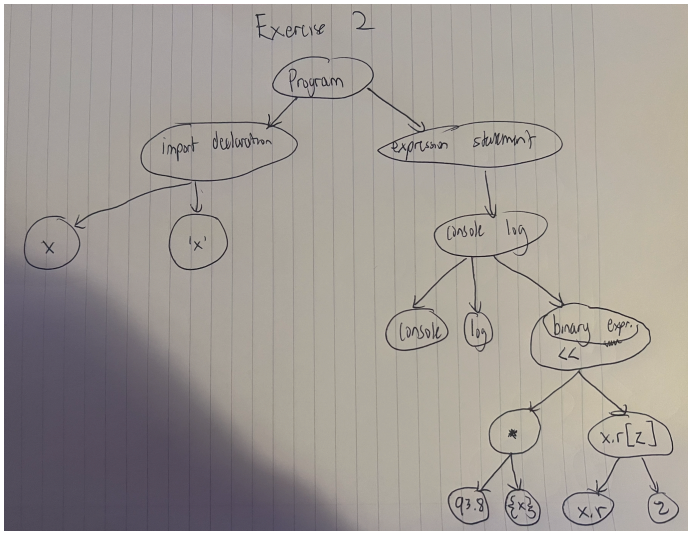
Affidavit: I, Patrick King, and Thomas Powell confirm that each of the readings and watchings were completed.

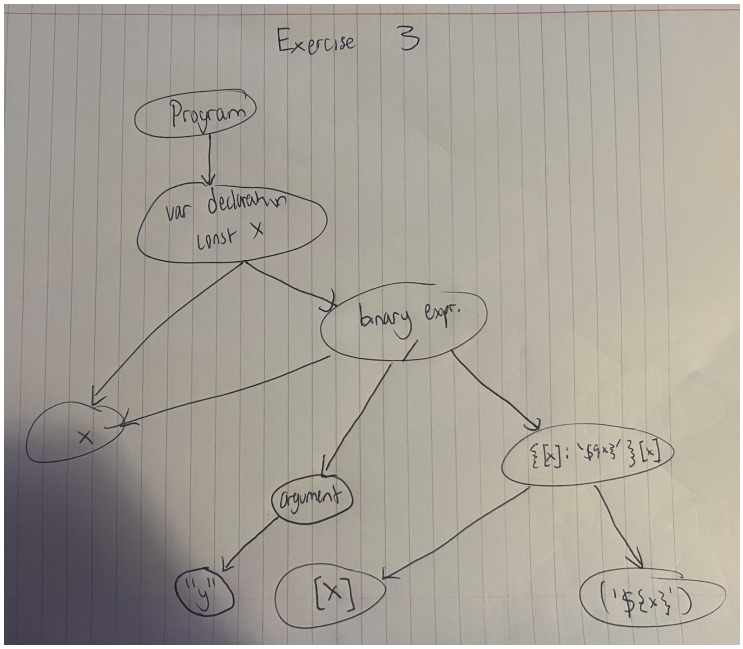
Exercises #1



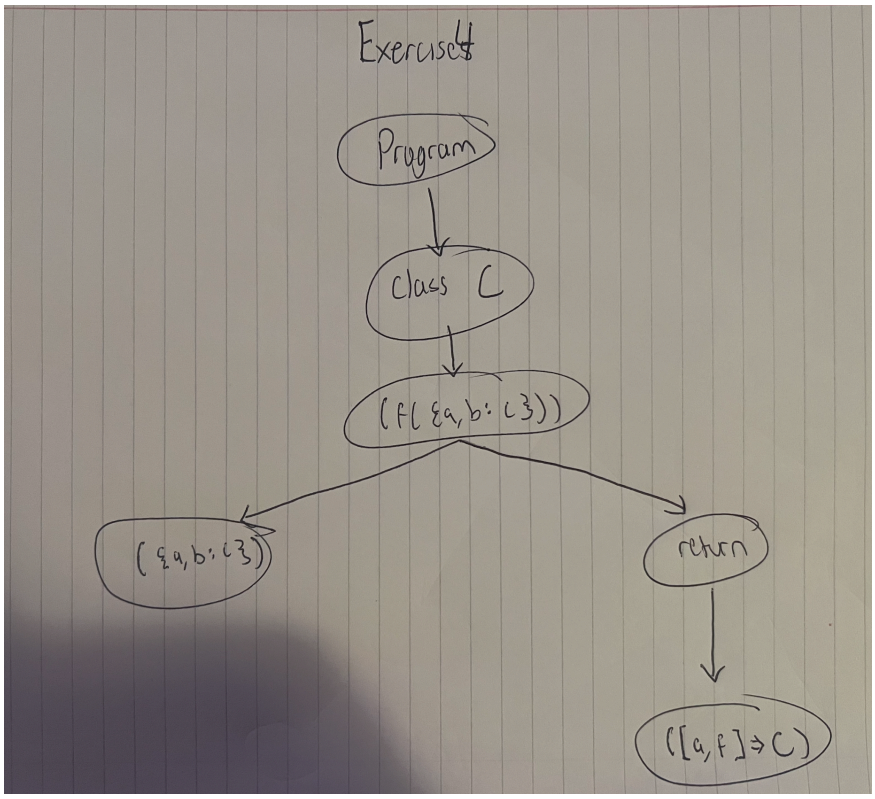
Exercise #2



Exercise #3



Exercise #4



Exercise #5a

```
Exp       = Term
Term      = Factor "**" Term
          | Factor
Factor    = "-" Factor
          | Primary
Primary   = number
          | "(" Exp ")"

number    = digit+
```

Exercise #5b

```
Exp       = Term
Term      = "-" Term
          | Factor "**" Term
          | Factor
Factor    = Primary
Primary   = number
          | "(" Exp ")"

number    = digit+
```

Exercise #5c

```
Exp       = Term
Term      = Factor "**" Term
          | Factor
Factor    = "-" Primary
          | Primary
Primary   = "(" Exp ")"
          | number

number    = digit+
```

Exercise #6

```
AstroPP {
  Program       = Statement+

  Statement
    = IfStmt
    | WhileStmt
    | BreakStmt
    | Assignment
    | Exp ";"

  IfStmt
```

```
    = "if" Exp Block ("else" (Block | IfStmt))?

  WhileStmt
    = "while" Exp Block

  BreakStmt
    = "break" ";"

  Assignment
    = id "=" Exp ";"

  Block
    = "{" Statement* "}"

  Exp
    = OrExp

  OrExp
    = AndExp ("||" AndExp)*

  AndExp
    = RelExp ("&&" RelExp)*

  RelExp
    = Primary (RelOp Primary)?

  RelOp
    = "==" | "!=" | ">" | "<" | ">=" | "<="

  Primary
    = id
    | number
    | "(" Exp ")"

  id        = letter+
  number    = digit+

  space    += "//" (~"\n" any)*
}
```

Project Repository:

https://github.com/patrickgking03/cloudscribe